

HW 6

17.2 2 Ω

1) some data $(x_i, z_i, y_i)_{i=1}^n$
 $\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 cov. binary outcome
 group indicator

At 2025: $\hat{y}_i = \hat{\gamma}_1 + x_i^T \hat{\beta}_1$ group 1

$$y_i = \beta_0 + x_i^T \beta_0 \quad \text{group 0}$$

joint: $y_i^1 = \hat{\alpha}_0 + \hat{\alpha}_z z_i + x_i^T \hat{\alpha}_x + z_i x_i^T \hat{\alpha}_{zx}$

1) $(\hat{\alpha}_0, \hat{\alpha}_z, \hat{\alpha}_x, \hat{\alpha}_{zx})$ in terms of $(\hat{\gamma}_1, \hat{\beta}_1, \hat{\gamma}_0, \hat{\beta}_0)$:

if $z_i = 0$

$$y_i = \alpha_0 + x_i^T \alpha = \alpha_0 + x_i^T \beta$$

if $z_i = 1$

$$y_i = \alpha_0 + \alpha_1 z + x_i^T \alpha_x + x_i^T \alpha_{zx} \\ = \alpha_0 + \alpha_1 z + x_i^T (\alpha_x + \alpha_{zx})$$

$$\alpha_0 = \gamma_0, \quad \alpha_X = \beta_0$$

$$\alpha_0 + \alpha_7 = \gamma_1$$

$$\alpha^1 x + \alpha^2 z x = \beta^1$$

$$\left(\alpha_0 = \beta_0 \right)$$

$$(\hat{\alpha}x = \hat{\beta}0$$

$$\hat{\alpha}_T = \hat{\gamma}_1 - \hat{\gamma}_0$$

$$\hat{\alpha} z x = \hat{\beta}_1 - \hat{\beta}_0$$

- 2) Prove fitted val of $\hat{y}_i$'s are same from the separate + pooled OLS for units $i=1, \dots, n$

(from 1):

case $z_i=0$: sep: $\hat{y}_{i \text{ sep}} = \hat{\gamma}_0 + x_i^T \hat{\beta}_1$ } $\hat{\gamma}_0 = \hat{\alpha}_0 + \hat{\alpha}_z z$
plug in $\hat{\beta}_1 = \hat{\alpha}_x + \hat{\alpha}_{zx}$
 $\hat{y}_{i \text{ sep}} = \hat{\alpha}_0 + \hat{\alpha}_z + x_i^T (\hat{\alpha}_x + \hat{\alpha}_{zx})$

" $\hat{y}_{i \text{ pooled}}$

case $z_i=1$: sep: $\hat{y}_{i \text{ sep}} = \hat{\gamma}_0 + x_i^T \hat{\beta}_0$ $\hat{\gamma}_0 = \hat{\alpha}_0$
 $\hat{\beta}_0 = \hat{\alpha}_x$

$$\hat{y}_{i \text{ sep}} = \hat{\alpha}_0 + x_i^T \hat{\alpha}_x = \hat{y}_{i \text{ pool}} \quad (\text{since } z_i=0)$$

3) $h_{ii} = [X(X^T X)^{-1} X^T]_{ii}$ for X the design matrix

19.4 Diff in-means w/ weights

2)

(from 19.3) the Galtonian formula for univ. wls

$$\min_{a,b} \sum_{i=1}^n w_i (y_i - a - bx_i)^2$$

has the minimizer $\hat{\beta}_w = \frac{\sum_{i=1}^n w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2}$

$$\hat{\alpha}_w = \bar{y}_w - \hat{\beta}_w \bar{x}_w$$

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad \bar{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i}$$

show x_i in wls of y_i on $(1, x_i)$ w/ weights $w_i (i=1, \dots, n)$ equals $\bar{y}_{w,1} - \bar{y}_{w,0}$

$$\bar{y}_{w,1} = \frac{\sum_{i=1}^n w_i x_i y_i}{\sum_{i=1}^n w_i x_i}$$

$$\bar{y}_{w,0} = \frac{\sum_{i=1}^n w_i (1-x_i) y_i}{\sum_{i=1}^n w_i (1-x_i)}$$

* binary variable $x_i \in \{0,1\}$

(num) $\hat{\beta}_w = \frac{\sum w_i x_i y_i - \bar{x}_w \sum w_i y_i - \bar{y}_w \sum w_i x_i + \bar{x}_w \bar{y}_w \sum w_i}{\sum w_i x_i (1-x_i)}$

$$= \sum w_i x_i y_i - \bar{x}_w (\sum w_i) \bar{y}_w \quad (?)$$

⑦

(denom) $\sum_{i=1}^n w_i (x_i - \bar{x}_w)^2 = \sum w_i x_i^2 - \frac{(\sum w_i)^2}{n} \bar{x}_w^2$

19.9 Page M. 19.2

3) WLS w/ $n \times 1$ vector Y
 $n \times K$ matrix X_1
 $n \times 1$ matrix X_2
 weights w_i 's

$$\text{WLS fit: } Y = X_1 \hat{\beta}_{w,1} + X_2 \hat{\beta}_{w,2} + \hat{\epsilon}_w$$

$\hat{\beta}_{w,2}$ equals coeff. $\tilde{X}_{w,2}$ in WLS fit of Y on $\tilde{X}_{w,2}$ where $\tilde{X}_{w,2}$ are residual vectors from column-wise WLS of X_2 on X_1

also eq. $\tilde{X}_{w,2}$ in WLS fit of $\tilde{Y}_w$ on $\tilde{X}_{w,2}$ where $\tilde{Y}_w$ residual vector from WLS on X_1

WLS objective is :

$$\min_{\beta_1, \beta_2} (Y - X_1 \beta_1 - X_2 \beta_2)^T W (Y - X_1 \beta_1 - X_2 \beta_2)$$

$$W = \text{diag}(w_1, \dots, w_n)$$

if mult. by $W^{1/2}$; just doing OLS basically
 $\hat{\beta}_{w,2}$ from WLS is $\hat{\beta}_2$ from OLS of $\tilde{Y}$ on $(\tilde{X}_1, \tilde{X}_2)$

(use FWL)

$$M_{\tilde{X}_1} \tilde{X}_2 = W^{1/2} [X_2 - X_1 (X_1^T W X_1)^{-1} X_1^T W X_2]$$

$\tilde{X}_{w,2}$

hw6

4.

a)

```
library(ISLR)
data(Auto)

mpg_median <- median(Auto$mpg)

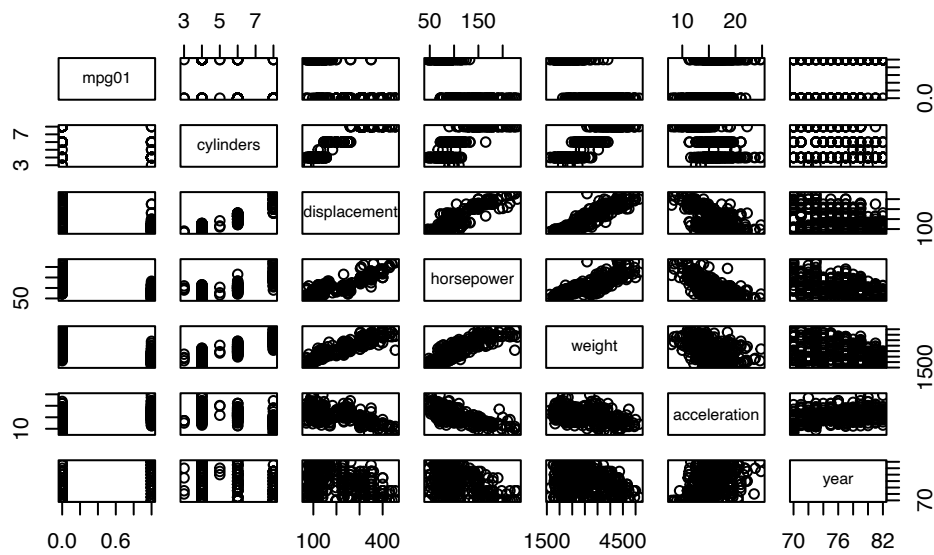
Auto$mpg01 <- ifelse(Auto$mpg > mpg_median, 1, 0)

table(Auto$mpg01)
```

```
  0    1
196 196
```

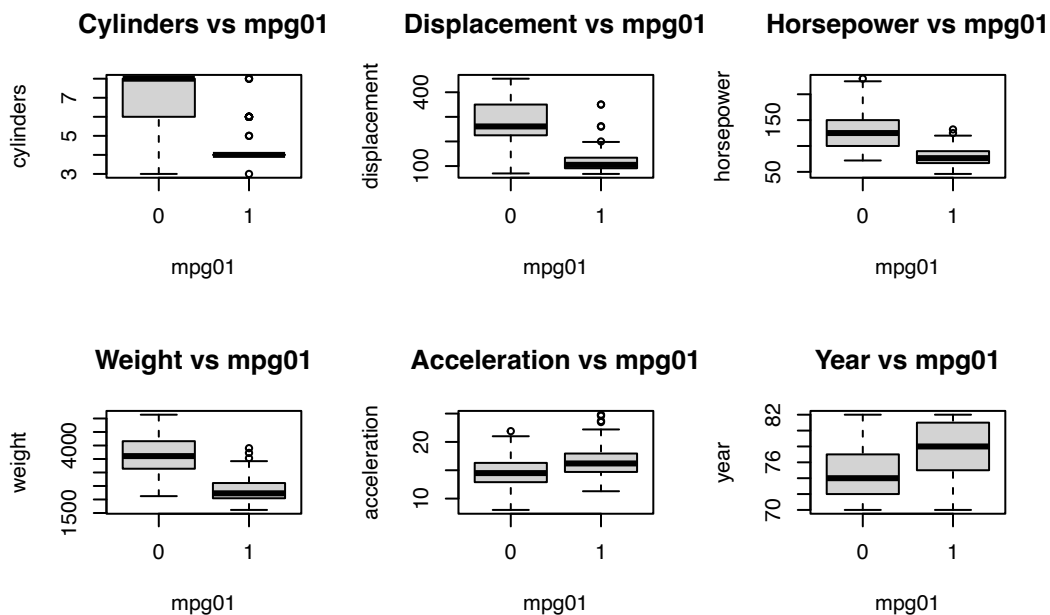
b)

```
pairs(Auto[, c("mpg01", "cylinders", "displacement", "horsepower", "weight", "acceleration",
```



```
par(mfrow = c(2, 3))

boxplot(cylinders ~ mpg01, data = Auto, main = "Cylinders vs mpg01")
boxplot(displacement ~ mpg01, data = Auto, main = "Displacement vs mpg01")
boxplot(horsepower ~ mpg01, data = Auto, main = "Horsepower vs mpg01")
boxplot(weight ~ mpg01, data = Auto, main = "Weight vs mpg01")
boxplot(acceleration ~ mpg01, data = Auto, main = "Acceleration vs mpg01")
boxplot(year ~ mpg01, data = Auto, main = "Year vs mpg01")
```



Displacement, horsepower, weight, and year seem like the most useful features based on the box plots? maybe?

c)

```
set.seed(1)

# Split data (70/30)
train_idx <- sample(1:nrow(Auto), 0.7 * nrow(Auto))

train <- Auto[train_idx, ]
test  <- Auto[-train_idx, ]

# Fit logistic regression
model_c <- glm(mpg01 ~ cylinders + displacement + horsepower + weight + year,
               data = train, family = binomial)

probs <- predict(model_c, newdata = test, type = "response")
preds <- ifelse(probs > 0.5, 1, 0)

test_error <- mean(preds != test$mpg01)
test_error
```

[1] 0.1016949

d)

```
#model_d <- glm(mpg01 ~ . - mpg - name, # remove mpg + name
#           data = train, family = binomial)

#probs_d <- predict(model_d, newdata = test, type = "response")
#preds_d <- ifelse(probs_d > 0.5, 1, 0)

#test_error_d <- mean(preds_d != test$mpg01)

#test_error
#test_error_d
```